



CHEM 103 A05 (51371)

Fall 2026

INTRO UNIVERSITY CHEMISTRY I

Territorial Acknowledgement

The University of Alberta, its buildings, labs and research stations are primarily located on the territory of the Néhiyaw (Cree), Niitsitapi (Blackfoot), Métis, Nakoda (Stoney), Dene, Haudenosaunee (Iroquois) and Anishinaabe (Ojibway/Saulteaux), lands that are now known as part of Treaties 6, 7 and 8 and homeland of the Métis. The University of Alberta respects the sovereignty, lands, histories, languages, knowledge systems and cultures of all First Nations, Métis, and Inuit nations.

Instructor Information

Instructor

Christie McDermott

Office: CHEM West W4-39C

Contact Information

cmcdermo@ualberta.ca

Office Hours: Tuesdays and Thursdays 10:00AM - 12:00PM

Lecture Room & Time

Tuesdays and Thursdays 3:30PM, CCIS 1-440

Laboratory Coordinator(s)

Coordinator Name: Dr. Yoram Apelblat

Email: yoram.apelblat@ualberta.ca

Office

CHEM East E2-34

Calendar Description

Atoms and molecules, states of matter, chemistry of the elements. Prerequisite: Chemistry 30, or equivalent. Note: Restricted to Engineering students only. Other students who take this course will receive 3 units.

Additional Course Information (General Content)

OVERVIEW

CHEM 103 is the first course of the introductory chemistry courses. The prerequisite is Chemistry 30 or equivalent. You are assumed to have knowledge of chemistry equivalent to the Alberta Chemistry 20–30 high school curriculum. You should prepare for this course by reviewing this prerequisite material. Students who do not have these prerequisites at the time of enrolment cannot expect supplementary professional tutoring from the instructor.

In this course, you will learn about the structure, bonding, and reactivity of chemical substances, focusing on the main-group elements. You will demonstrate your understanding of chemistry by visualizing chemical models from an atomic and molecular point of view, thinking logically to solve chemical problems and applying these ideas and skills to the real world. Hopefully, you will gain an appreciation of the influence of chemistry in your life and you will be able to think critically about chemical issues. In the lecture portion of the course, we will work on many specific learning outcomes (see next section for examples).

Throughout the course, problem sets will be essential so that you will build problem-solving strategies. By the end of the course, you should be able to apply concepts learned throughout the term to solve new types of problems (perhaps some you haven't seen before).

LABORATORY

You must register in, attend, and complete the laboratory portion of the course. The lab and course Canvas websites are separate. Read the instructions in the manual before the first lab.

HELP SESSIONS (Seminars)

These are equivalent to the seminar in which you have registered; they are not structured classes but rather tutorial help sessions in which you can consult teaching assistants for help with both lecture and lab topics. You would only attend if you need help and at any time no matter what seminar time you registered in. Please refer to the lab (Canvas) website for the schedule.

ADMINISTRATION

Questions about lectures, exams, or problem sets? Attend office hours and/or contact your instructor.

Questions about registration changes? Contact [Chemistry Undergraduate Student Services](#)

Questions about lab scheduling? Contact Dr. Yoram Apelblat (apelblat@ualberta.ca).

Help with lab reports and lecture topics? Consult TAs during help sessions.

Course Objectives/Outcomes

- Analyze one-electron systems using the Bohr model of atomic structure and some basic quantum principles such as: Heisenberg's uncertainty principle, dual nature of light and matter, wave functions, probability distributions, quantum numbers.
- Correlate quantum numbers (n , l , m_l , m_s) to atomic orbital energies, shapes and spatial orientations.
- Differentiate between different types of bonding, including covalent, ionic, and metallic bonding.
- For ideal and real gases, predict and calculate properties (pressure, volume, temperature, moles, molar mass) for pure and mixed gas systems.
- Generate Lewis structures for covalent molecules and ions and, coupled with VSEPR and Valence Bond Theory, determine the molecule's geometry, polarity, and bonding (bond order, bond length, strength and energy).
- In many-electron systems, predict atomic and ionic ground-state electron configurations and construct electron energy diagrams.
- Predict physical properties for ionic and metallic solids and use crystal lattice systems to calculate density, atomic radius, and packing efficiency.
- Use models of atomic structure to predict trends in reactivity and in periodic properties such as atomic and ionic radii, ionization energy, electron affinity.
- Use Molecular Orbital Theory to assess the bonding (bond order and

strength) and magnetic properties in second-row homonuclear diatomic molecules and ions.

- Use molecular structures and intermolecular forces to predict physical properties (boiling and melting points, vapour pressure, viscosity, capillarity, solubility) for liquids, solids, gases, and solutions.
- Utilize different CHEM 103 learning outcomes to predict reactivity and physical properties of main group elements and compounds.

Required Course Materials

Access to Stemble - our online resource and quiz platform.

Link: <https://www.stemble.com/>

Introductory University Chemistry I Laboratory Manual (Chem 101/103)

Edition: 2026-2027 Ed.

Available in CHEM West W1-50

Laboratory Personal Protection items: safety glasses, lab coat, covered footwear, long pants.

Lab coats and safety glasses available for purchase at the bookstore or CHEM West W1-50.

Textbook: Chemistry 2e (OpenStax)

Authors: Paul Flowers, Klaus Theopold, Richard Langley, William R. Robinson

Link: <https://openstax.org/details/books/chemistry-2e>

Note: this textbook is freely available online and can be accessed through the above website:

Grade Evaluation

Title	Assessment Type	Weight	Date of Evalu...
Online Quizzes (Stemble)	Quiz	5%	
Midterm Exam 1 (9:30 - 10:30AM)	Midterm	10%	10/3/26

Title	Assessment Type	Weight	Date of Evalu...
Midterm Exam 2 (9:00 - 11:00AM)	Midterm	20%	10/31/26
Laboratory	Labwork	25%	
Final Exam (Schedule TBA-- students should be available during the entire exam period: Dec. 9 - 22, 2026)	Exam (final)	40%	

For courses with a Final Exam, students must verify the date of the Final Exam on Bear Tracks when the Final Exam Schedule is posted.

Grades are unofficial until approved by the Department and/or Faculty offering the course.

Exam Conduct

Please refer to the [Examinations](#) section of the Academic Calendar for more details on Conduct of Exams.

Some key points to be aware of:

- All exams are closed book and cumulative. In accordance with Faculty of Science policy, only [Engineering-approved, non-programmable calculators](#) are permitted in exams; all other electronic devices are not permitted. You are permitted to use only those materials specified in the exam description; a data and formula sheet is provided. Students who use unauthorized materials will be in violation of the Academic Integrity Policy.
- Students must arrive at the specified time to take the exam. Once the exam has started, students must remain in the physical in-person or remote environment for at least 30 minutes. Students who arrive more than 30 minutes late for an in-person exam will not be permitted to take the exam.
- All cell phones must be turned off and stored in your bags.
- Representative questions from past exams will be available on Canvas.
- Your student photo ID is required to verify your identity.

Access to Past or Representative Evaluative Material

Representative material from past exams will be available on Canvas the week prior to the exam.

Final Letter Grades

University of Alberta Grading Policy

Grades reflect judgments of student achievement made by instructors and must correspond to the associated descriptor. These judgements are based on a combination of absolute achievement and relative performance in a class. Faculties may define acceptable grading practices in their disciplines. Such grading practices must align with the University of Alberta [Assessment and Grading Policy](#) and its procedures, which are available online at the [University of Alberta Policies and Procedures Online \(UAPPOL\)](#) website.

Course Grades Obtained by Undergraduate Students

Descriptor	Letter Grade	Grade Point Value
Excellent	A+	4.0
Excellent	A	4.0
Excellent	A-	3.7
Good	B+	3.3
Good	B	3.0
Good	B-	2.7
Satisfactory	C+	2.3
Satisfactory	C	2.0
Satisfactory	C-	1.7
Poor	D+	1.3
Minimal Pass	D	1.0
Failure	F or F4	0.0
Note: F4 denotes the eligibility of a student to apply for a re-examination of a course.		

Policies for Late and Missed Term Work

ABSENCES

A student who cannot complete Midterm Exam 1 or 2 due to incapacitating illness, severe domestic affliction, or other compelling reasons must contact the instructor within two working days of missing exam, or as soon as possible, to request an excused absence. If an excused absence is granted, then the weight of the missed exam will be transferred to the final exam without exception. An excused absence is a privilege and not a right. There is no guarantee that an absence will be excused. Any absence that is unexcused will result in a mark of zero. Misrepresentation of facts to gain an excused absence is a serious breach of the [Student Academic Integrity Policy](#). In all cases, instructors may request adequate documentation to substantiate the reason for the absence, at their discretion, such as a [statutory declaration](#). Students who have schedule conflicts with a midterm exam (due to scheduled classes, labs, or seminars) must inform me within **two weeks of the start of classes**.

For online quizzes, no late submissions or "absences" will be accepted.

Deferred Final Examination

A student who cannot write the final examination due to incapacitating illness, severe domestic affliction, or other compelling reasons can apply for a deferred final examination. Such an application must be made to the student's home Faculty Office within two working days of the missed exam and must be supported by appropriate documentation or a Statutory Declaration (see calendar on [Attendance](#)). Deferred examinations are a privilege and not a right; there is no guarantee that a deferred examination will be granted. The Faculty may deny deferral requests in cases where less than 50% of term work has been completed. Misrepresentation of facts to gain a deferred examination is a serious breach of the [Student Academic Integrity Policy](#). The deferred examination for CHEM 103 is scheduled for Tuesday, January 5th, 1:00 - 4:00PM, location TBA.

Missed Term Work/Final Exam Due to Non-medical Protected Grounds (e.g., religious beliefs)

When a term assessment or final exam presents a conflict based on [non-medical protected grounds](#), students can register with the Academic Success Centre for accommodations via their [Register for Accommodations](#) website. Students can review their eligibility and choose the registration process specific for **Accommodations Based on Non-medical Protected Grounds**.

It is imperative that students review the dates of all course assessments upon receipt of the course syllabus, and register **AS SOON AS POSSIBLE** to ensure the timely application of the accommodation. Students who register later in the term may experience unavoidable delays in the processing of the registration, which can affect the accommodation.

Academic Integrity and Student Conduct

The University of Alberta is committed to the highest standards of academic integrity and honesty. Students are expected to be familiar with these standards regarding academic honesty and to uphold the policies of the University in this respect. Students are particularly urged to familiarize themselves with the provisions of the [Student Academic Integrity Policy](#) (on the [University of Alberta Policies and Procedures Online \(UAPPOL\) website](#)) and avoid any behaviour which could potentially result in suspicions of cheating, plagiarism, misrepresentation, and/or unauthorised collaboration. Academic dishonesty is a serious offence and can result in suspension or expulsion from the University.

The University of Alberta is also committed to maintaining a learning environment that fosters the safety, security, and the inherent dignity of each member of the community, ensuring students conduct themselves accordingly by avoiding behaviour stipulated in the [Student Conduct Policy](#) (e.g., discrimination, harassment, physical assault).

All students are expected to consult the [Academic Integrity website](#) for clarification on the various academic offences. All forms of academic dishonesty are unacceptable at the University. Unfamiliarity of the rules, procrastination or personal pressures are not acceptable excuses for committing an offence. Listen to your instructor, be a good person, ask for help when you need it, and do your own work -- this will lead you toward a path to success. Any academic integrity concern in this course will be reported to the College of Natural and Applied Sciences. Suspected cases of non-academic misconduct will be reported to the Office of Student Success and Experience (formerly the Office of the Dean of Students). The College, Faculty, and Dean of Students are committed to student rights and responsibilities, and adhere to due process and administrative fairness, as outlined in the [Student Academic Integrity Policy](#) and the [Student Conduct Policy](#). Please refer to the policies for details on inappropriate behaviours and possible sanctions.

The College of Natural and Applied Sciences (CNAS) has created an [Academic Integrity for CNAS Students](#) website. Website content includes the importance of academic integrity, examples of academic misconduct & possible sanctions, and the academic misconduct & appeal process. Students can also access this material as an [online, self-directed Canvas course](#) and complete assessments to test their knowledge.

The Office of Student Conduct and Accountability has also made the following Canvas course available for all students: [Academic Citizenship: Integrity and Belonging in the Learning Community](#).

"Integrity is doing the right thing, even when no one is watching" -- C.S. Lewis

Remote Delivery Considerations

Recordings of Synchronous Activities:

- Please note that class times for this course will be recorded. Recordings of this course will be used for the purposes of synchronous remote lecture attendance in case of illness, and review of lecture content. Recordings will be shared with students enrolled in this section of CHEM 103 and any

- instructors who may have access to our course Canvas site.
- Students have the right to not participate in the recording and, if watching remotely, are advised to remove all identifiable and personal belongings from the space in which they will be participating and keep their cameras and audio off. Please note that the chat will not be monitored during lectures.
 - Lecture recordings will be made available until the final exam and will be accessible by Canvas links.
 - Any other recordings done by students in this course, audio or video recording, digital, or otherwise, of lectures, labs, seminars, or any other teaching environment, is allowed only with prior written consent or as part of an approved accommodation plan,. Student or instructor content, digital or otherwise, created or use within the context of the course, is to be used solely for personal study and not to be used or distributed without prior consent from the content authors.

AI Assisted Grading

This course uses AI-assisted tools—specifically *Stemble*—to support parts of the grading process. These tools assist in identifying patterns, enhancing consistency, and the efficiency of grading tasks. By streamlining these processes, more time can be dedicated to higher-value teaching activities, such as providing personalized feedback, developing learning materials, or directly supporting students, while also ensuring timely return of assessments. AI tools are used to assist in administering and grading online quizzes. Regardless of the tool's involvement, final responsibility for all grades rests with the instructor.

Data Use and Privacy:

Student submissions may be processed through these tools to provide grading support. In alignment with the tools terms of use:

- Your work is **not used to train public AI models**.
- Submitted data is typically **retained temporarily** to complete the grading task but is **not stored long-term**.
- No personally identifiable information is intentionally shared with the AI tool.

Stemble has been reviewed through the University's Privacy and Security Assessment process and is approved for use in AI-assisted grading. If you are concerned about how your data is handled, I encourage you to reach out to me with questions or review the tool's privacy documentation. The appeals process for grades remains unchanged and follows standard university procedures.

Appeals & Error Handling:

If you believe your grade reflects an oversight—by the AI tool or a human reviewer, please review the rubric and contact me within one week for re-evaluation.

On-Line Homework Disclaimer

Online quizzes are a component of this course and are provided by a third-party company (Stemble). Please be aware that this company will be storing assessment information that may be associated with you. As a way to protect your personal information, you may be assigned a random ID to enter into the system. Only the random ID, performance on the online homework, and the affiliation to the University of Alberta for this random ID will be conveyed to the company. You are not required to provide any additional personal information to this company. If you have concerns about this, please contact the instructor of the course.

Students who wish to opt out of the use of the fee-for-service Stemble system have the option to utilize the library computers which have been loaded with the necessary software.

Students choosing to access and use the online resources should note the following:

1. Registration in the system and any monetary transactions are of their own accord and not the responsibility of the University of Alberta.
2. Students should be mindful of protecting personal information and aware of how their personal information might be used and/or shared.
3. Students can consider using a non-identifying email address or account for these purposes.

Statement of Expectation for AI Use

This course fosters the knowledge, critical, and affective skills that are essential to your development. To achieve the course learning outcomes, you will need to

practice and demonstrate these skills **without substantive assistance** from others or GenAI.

Important: You are welcome to use AI tools as **personal learning supports**. However AI use is **prohibited in assessed work, including LABS, graded quizzes and exams**. Failure to abide by this guideline may be considered an act of cheating and a violation as outlined in the relevant sections of the University of Alberta [Student Academic Integrity Policy](#).

Contract Cheating and Misuse of University Academic Materials or Other Assets

Contract cheating describes the form of **academic dishonesty** where students get academic work completed on their behalf, which they submit for academic credit as if they had created it themselves. Contract cheating may or may not involve the payment of a fee to a third party, who then creates the work for the student.

Examples include:

1. Getting someone to write an essay or research paper for you.
2. Getting someone to complete your assignment or exam for you.
3. Posting an essay, assignment, or exam question to a tutorial or study website; the question is answered by a "content expert", then you copy it and submit it as your own answer.
4. Posting your solutions to a tutorial/study website, public server, or group chat and/or copying solutions that were posted to a tutorial/study website, public server, or group chat.
5. Sharing your login credentials to the course management system (e.g., Canvas) and allowing someone else to complete your assignment or exam remotely.
6. Using an artificial intelligence bot or text generator tool to complete your essay, research paper, assignment, or exam solutions for you (without the instructor's permission).
7. Using an online grammar checker to "fix" your essay, research paper, assignment, or exam solutions for you (without the instructor's permission).

Contract cheating companies thrive on making students believe that they cannot succeed without their help; they attempt to convince students that cheating is the

only way to succeed.

Uploading the instructor's teaching materials (e.g., course outlines, lecture slides, assignment, or exam questions, etc.) to tutorial, study or note-sharing websites, public servers, or chat apps is a copyright infringement and constitutes the **misuse of University academic materials or other assets**. Receiving assignment solutions or answers to exam questions from an unauthorized source puts you at risk of receiving inaccurate information.

Appropriate Collaboration

Students need to be able to recognize when they have crossed the line between appropriate collaboration and inappropriate collaboration. If students are unsure, they need to ask instructors to clarify what is allowed and what is not allowed.

Here are some tips to avoid copying on assessments:

1. Do not write down something that you cannot explain to your instructor.
2. When you are helping other students, avoid showing them your work directly. Instead, explain your solution verbally. Allowing your work to be copied is also considered inappropriate collaboration.
3. It is also possible that verbally discussing the solution in too much detail may result in written responses that are too similar. Try to keep discussions at a general or higher level.
4. If you find yourself reading another student's solution, do not write anything down. Once you understand how to solve the problem, remove the other person's work from your sight and then write up the solution to the question yourself. Looking back and forth between someone else's paper and your own paper is almost certainly copying and considered inappropriate collaboration.
5. If the instructor or TA writes down part of a solution in order to help explain it to you or the class, you cannot copy it and hand it in for credit. Treat it the same way you would treat another student's work with respect to copying, that is, remove the explanation from your sight and then write up the solution yourself.
6. There is often more than one way to solve a problem. Choose the method that makes the most sense to you rather than the method that other students happen to use. If none of the ideas in your solution are your own, there is a good chance it will be flagged as copying.

How to Succeed in this Course

HOW TO SUCCEED IN THIS COURSE

Learning chemistry requires work and commitment. To succeed in this course, you should:

1. **Attend lectures and take notes.** The lectures define the scope of the material for which you are responsible. Students who regularly miss lectures do not fare well in the course. Take notes in class to reinforce your understanding of the material. Resist the urge to check your phone or social media. Continue to pay attention even when you are not writing notes - you are also responsible for information that may not be explicitly written down. Be brave and ask questions in class. If you miss a lecture, review recordings provided on eClass. I record my lectures as a courtesy to students who have to occasionally miss a class for illness or other reasons. To encourage you to attend class, I will be embedding response questions in many of the lectures. These response questions will be available only for a limited time, during class, and student answers will be recorded in the grade book on Canvas as a way to track attendance. After the fourth class meeting (after September 10th), I will begin recording students' attendance via this method.
2. **Complete online activities, lessons, and problem sets; use your textbook; and do your homework.** Before lectures, read the assigned sections in the textbook. Review the sample problems, and try the recommended end-of-chapter problems, ensuring that you can complete each learning objective confidently. Complete the problem sets; although they are not marked, they give you a good idea of the types of questions expected on exams and serve as excellent preparation for exams. Online quizzes will be given regularly to test your learning.
3. **Take responsibility for your own learning.** In a large university setting, you control your own learning; no one is supervising you. Engage actively in learning and seek out the numerous resources available to you: TAs in help sessions, your instructor during office hours, the textbook package and its online resources, and various university services. The [Academic Success Centre](#) provides professional academic support to help students strengthen their skills and achieve their goals. If you are eligible and registered for [accessibility accommodations](#), you are responsible for adhering to procedures and deadlines.

University Policy

Withdrawals

See the [University Calendar](#) for the relevant [add/drop deadlines](#) for each term.

Course Outlines

Policy about course outlines can be found in the [Academic Regulations, Evaluation Procedures and Grading section](#) of the University Calendar.

Student Academic Integrity

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Recordings

Audio or video recording, digital or otherwise, of lectures, labs, seminars or any other teaching environment by students is allowed only with the prior written consent of the instructor or as a part of an approved accommodation plan. Student or instructor content, digital or otherwise, created and/or used within the context of the course is to be used solely for personal study, and is not to be used or distributed for any other purpose without prior written consent from the content authors.

Accommodations for Students

In accordance with the University of Alberta's [Accommodation Policy](#) and [Discrimination and Harassment Policy](#), accommodation support is available to eligible students who encounter limitations or restrictions to their ability to perform the daily activities necessary to pursue studies at a post-secondary level due to medical conditions and/or non-medical protected grounds. Accommodations are coordinated through the [Academic Success Centre](#), and students can learn more about eligibility on the [Register for Accommodations website](#).

It is recommended that students apply AS SOON AS POSSIBLE in order to ensure sufficient time to complete accommodation registration and coordination. Students are advised to review and adhere to published deadlines for accommodation approval and for specific accommodation requests (e.g., exam registration submission deadlines). Students who request accommodations less than a month in advance of the academic term for which they require accommodations may experience unavoidable delays or consequences in their academic programs, and may need to consider alternative academic schedules.

Course Material Access

[Course Material Access](#) allows students to access their required course materials through Canvas for the duration of the term. A flat fee per term will appear in the student's Fee Assessment section in Bear Tracks. Digital formats are prioritized, but print materials are provided if digital formats are not available. **This is an automatic opt in program, but students can choose to opt out**, if they decide it is not for them. Some reasons may include:

- the cost of all required course material is less than the program flat fee for the term
- print materials are preferred over digital formats
- access to course material is desired beyond a given term

Students must choose to opt out by each term's [registration deadline \(i.e., add/delete date\)](#). Students will automatically be opted out if none of their digital or printed course material has a cost associated with it. For more information on

how to access your course materials on Canvas or how to opt out, visit the [Course Material Access](#) website.

Learning and Working Environment

The Faculty of Science is committed to ensuring that all students, faculty and staff are able to work and study in an environment that is safe and free from discrimination, harassment, and violence of any kind. It does not tolerate behaviour that undermines that environment. This includes virtual environments and platforms.

If you are experiencing harassment, discrimination, fraud, theft or any other issue and would like to get confidential advice, please contact any of these campus services:

- [Office of Safe Disclosure & Human Rights](#): A safe, neutral and confidential space to disclose concerns about how the University of Alberta policies, procedures or ethical standards are being applied. They provide strategic advice and referral on matters such as discrimination, harassment, duty to accommodate and wrong-doings. Disclosures can be made in person or online using the [Online Reporting Tool](#).
- [University of Alberta Protective Services](#): Peace officers dedicated to ensuring the safety and security of U of A campuses and community. Staff or students can contact UAPS to make a report if they feel unsafe, threatened, or targeted on campus or by another member of the university community.
- [Office of the Student Ombuds](#): A confidential and free service that strives to ensure that university processes related to students operate as fairly as possible. They offer information, advice, and support to students, faculty, and staff as they deal with academic, discipline, interpersonal, and financial issues related to student programs.
- [Office of Student Success and Experience](#): They can assist students in navigating services to ensure they receive appropriate and timely resources. For students who are unsure of the support they may need, are concerned about how to access services on campus, or feel like they may need interim support while they wait to access a service, this office is there to help.

Student Supports

Faculty of Science Student Services: The [Faculty of Science Student Services](#) office is located on the main floor of the Centennial Centre for Interdisciplinary Sciences (CCIS). This office can assist with the planning of [Your Academics](#), and provide information related to [Student Life & Engagement](#), [Internship and Careers](#), and [Study Abroad](#) opportunities. Please visit [Advising](#) for more information about what Faculty Academic Advisors can assist you with.

Academic Success Centre: The [Academic Success Centre](#) provides professional academic support to help students strengthen their academic skills and achieve their academic goals. Individual advising, appointments, and group workshops are available year round in the areas of Accessibility, Communication, Learning, and Writing Resources. Modest fees may apply for some services.

Feeling Stressed, Anxious, or Upset? It's normal for us to have different mental health experiences throughout the year. Know that there are people who want to help. You can reach out to your friends and access a variety of supports available on and off campus at the [Need Help Now](#) webpage or by calling the 24-hour Distress Line: 780-482-4357 (HELP).

Disclaimer

Any typographical errors in this syllabus are subject to change and will be announced in class and/or posted on the course website. The date of final examinations is set by the Registrar and takes precedence over the final examination date reported in the syllabus.

LECTURE SCHEDULE — CHEM 103 SECTION A05 — FALL (SEPT–DEC) 2026

Week	Topics	Reading *
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Tues. Sept. 1	UNIT 1. ATOMS <i>Review: Atomic theory, elements and compounds</i>	Chaps. 2, 3
Thurs. Sept. 3	Nature of light, atomic spectra	6.1
Tues. Sept. 8	Bohr model, wave-particle duality, quantum mechanics	6.2–6.3
Thurs. Sept. 10	Hydrogen orbitals, many-electron atoms	6.4
Tues. Sept. 15	Shielding, electron configurations	6.4
Thurs. Sept. 17	Ions, periodic table, trends in atomic properties	6.5
Tues. Sept. 22	Trends in chemical properties; hydrogen	18.1, 18.5
Thurs. Sept. 24	Alkali metals UNIT 2. MOLECULES <i>Review: Ionic, covalent, and metallic bonding</i>	18.2 7.1–7.2
Tues. Sept. 29	Covalent bond strengths and energy; electronegativity, bond polarity	7.5–7.6
Thurs. Oct. 1	Lewis structures	7.3–7.4
Sat. Oct. 3	MIDTERM EXAM 1, 9:30 am – 10:30 am	
Tues. Oct. 6	VSEPR and molecular shapes	7.6
Thurs. Oct. 8	Valence bond theory and orbital hybridization	8.1–8.3
Tues. Oct. 13	Molecular orbital theory and electron delocalization	8.4
Thurs. Oct. 15	Chemistry of selected main-group elements	18.6–18.11
Tues. Oct. 20	Chemistry of selected main-group elements	18.6–18.11
Thurs. Oct. 22	Chemistry of selected main-group elements	18.6–18.11
Tues. Oct. 27	Acids and bases – definitions	14.1
Thurs. Oct. 29	Acids and bases – structure and strengths	14.3
Sat. Oct. 31	MIDTERM EXAM 2, 9:00 am – 11:00 am	

Tues. Nov. 3	UNIT 3. GASES, LIQUIDS, AND SOLIDS <i>Review: Gases, ideal gas law, mixtures of gases</i>	9.1–9.3
Thurs. Nov. 5	Kinetic molecular theory, real gases	9.5–9.6
Tues. Nov. 10	READING WEEK (NO CLASS)	
Thurs. Nov. 12	READING WEEK (NO CLASS)	
Tues. Nov. 17	Phase changes	10.3
Thurs. Nov. 19	Phase diagrams, types of intermolecular forces	10.1, 10.4
Tues. Nov. 24	Liquids and solubility	10.2
Thurs. Nov. 26	Crystalline solids	10.5–10.6
Tues. Dec. 1	Structures of metallic and ionic solids	10.6
Thurs. Dec. 3	Advanced materials; carbon and silicon	18.3–18.4
Tues. Dec. 8	Review	

[Dec. exam period](#) [FINAL EXAM \(schedule to be announced\)](#)

* Keyed to Flowers et al., *Chemistry 2e*.

Copyright

Dr. C. McDermott Department of Chemistry, Faculty of Science, University of Alberta (2026)